

Task 25 – Multi-Table Sales Analysis

Northwind Traders | SQL + Power BI

Objective

Combine orders, products and customers using relational joins, validate relationships, prevent double counting and present business insights in Power BI.

Dataset & Relationships

The analysis uses the Microsoft Northwind relational sample. The core analytical path is Customers → Orders → Order Details → Products → Categories, with Employees and Shippers connected to Orders.

Reference KPI Snapshot

Metric	Reference value
Orders	830
Customers	91
Products	77
Units sold	51,317
Gross line-item sales	\$1,354,459

Category Performance

Category	Gross sales
Beverages	\$286,527
Dairy Products	\$251,330
Meat/Poultry	\$178,188
Confections	\$177,099
Seafood	\$141,624
Condiments	\$113,695
Produce	\$105,269
Grains/Cereals	\$100,727

Five Insights

- Beverages has the highest gross sales among the eight Northwind categories.
- Dairy Products is the second-largest category and is supported by several high-volume products.
- Côte de Blaye is a major revenue driver because of its high unit price.
- Customer and country dimensions enable drill-down from overall sales to market contribution.
- Freight must be calculated at order level; directly summing Orders.Freight after joining Order Details can inflate the result.

Double-Counting Control

Orders and Order Details have a one-to-many relationship. A single order can contain several detail rows. Revenue belongs at the line-item level, while Freight belongs at the order level. The SQL deliverable therefore calculates NetSales from Order Details and handles freight in separate order-level queries/CTEs.

Power BI Dashboard Design

Recommended visuals: KPI cards for Total Sales, Orders, Customers, Units and Average Order Value; a monthly sales trend; category sales; Top 10 products; Top 10 customers; country sales; and shipping performance.

Submission note

The numerical KPI snapshot is a reference based on publicly documented Northwind analyses. Run the supplied SQL against the exact Northwind copy used for the internship before presenting the figures as final measured results.